SOCRATES Phase 2, Stage 2:

Poly-Algebraic Calculus — Hypergraph Dimension Estimation,
a Ten-Problem Physics Benchmark, and Comparison to Classical Methods

Xavier Callens
Socrate AI Lab

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Abstract

This report documents Phase 2 (Stage 2) of the Socrate AI Lab programme: the construction and rigorous benchmarking of a computational "Poly-Algebraic Calculus" toolkit for hypergraph-based physics, following Wolfram's proposal that space, time, and known physics may emerge from discrete rewriting of a pre-geometric hypergraph. Of four brainstormed formalization concepts, two were selected for implementation on tractability grounds: a higher-arity rule-search calculus (*Poly-Algebraic Calculus* proper) and an emergent-dimension estimator (*Pre-Geometric Dimension Variables*). The dimension estimator was applied to ten known physics problems with independently verifiable dimension targets — exact results for closed orbits and rigorous theorems, and literature values (Grassberger–Procaccia, 1983) for two classical chaotic attractors. The initial benchmark run produced a headline of "10/10 passed" that a subsequent audit showed to be substantially misleading: six of the ten passes came from a mathematically degenerate regime, and neither chaotic result was converged in point count. Two independent, real software defects were found in the process and are corrected here, with regression tests and measured before/after evidence, including a $\sim 27\times$ performance improvement. A genuine classical baseline (the Grassberger–Procaccia correlation-sum method) and an apples-to-apples comparison harness were then built so that "better than the traditional approach" could be a checkable claim rather than an assertion. A systematic re-comparison across all ten problems, targeting a majority win rate against this baseline, was carried out and passed through three independent review passes (self-report, audit, and an adversarial skeptic checking the audit itself). The programme's goal of at least eight of ten problems was not met: the final, three-times-verified count is four of ten, all four sharing one underlying mechanism (a smooth closed one-dimensional orbit's ring-lattice degeneracy), and zero of four density-robustness claims survived a proper control. The strongest evidence of a genuine estimator advantage — high accuracy on two non-degenerate two-dimensional targets where the traditional method never converges — falls outside what the stated win criterion can score and is reported as the recommended focus for the next round.

Contents

1 Introduction and Motivation

Stephen Wolfram's Physics Project proposes that space, time, and the known laws of physics are not fundamental, but emerge from the discrete, repeated application of local rewriting rules to a pre-geometric hypergraph. Formalizing this hypothesis computationally requires new mathematical and software tooling, analogous to how historical paradigm shifts in physics required new mathematical syntax (algebra for the unknown quantity, calculus for instantaneous change, complex numbers for solutions "off the real line").

Four concepts were brainstormed as candidate formal tools for this programme:

1. **Poly-Algebraic Calculus** (higher-arity algebra): extending algebra's "solve for an unknown quantity" to "solve for an unknown higher-arity relation or rewrite rule."

2. **Branchial Cohomology**: a topological invariant of the multiway (branching) graph of all possible rewrite histories, proposed as the discrete analogue of quantum interference.

3. **Rulial Operators**: discrete derivatives measuring the rate of topological deformation per rewrite step, proposed as a discrete analogue of energy/mass density.

4. **Pre-Geometric Dimension Variables**: treating the dimension of emergent space as a fluid, locally-varying quantity inferred from how the hypergraph's ball volume scales with radius.

1.1 Selection rationale

Concepts #1 and #4 were selected for implementation. Concept #4 (dimension estimation via volume-growth scaling) is a well-established technique already used in causal set theory and in Wolfram's own work — implementable immediately with no new mathematics required. Concept #1 (a rule-search calculus) is the substrate every other concept implicitly depends on: one cannot measure a dimension, a rulial derivative, or a branchial invariant without first having a rigorous language for hypergraphs, patterns, and rewrites. Concepts #2 and #3 were assessed as the least tractable: *Branchial Cohomology* in particular has no working mathematical definition yet beyond an evocative name — no chain complex, no boundary operator has been specified for it anywhere in the literature this programme is aware of, and implementing it would have required inventing the mathematics from scratch rather than assembling existing tools, in tension with this programme's discipline of not asserting more than a computation actually proves.

2 Methods

2.1 Hypergraph representation and rewriting

A hypergraph is represented as an immutable multiset of ordered hyperedges over an implicit node set (nodes exist only by virtue of participating in a relation, following the Wolfram Model convention). Rewrite rules are specified as a left-hand-side pattern (a tuple of pattern edges over pattern variables) and a right-hand-side replacement; matching is performed by backtracking search over consistent pattern-variable assignments (subhypergraph pattern matching). Two evolution modes are implemented: a single-history mode that applies all non-overlapping matches per generation (the standard Wolfram Model update rule), and a multiway mode that branches over every individual single-match update, producing the full branching-state graph of the abstract rewriting system.

2.2 Emergent dimension estimation

For a hypergraph flattened into its adjacency graph (each hyperedge becomes a clique on its constituent nodes), the ball of radius r around a node is the set of nodes reachable within r hops. If the graph behaves like a d-dimensional lattice at some scale, ball volume grows as $|B(r)| \sim r^d$. The estimator fits *shell size* ($dV(r) = |B(r)| - |B(r-1)|$), not cumulative volume, against radius in log-log space: cumulative volume on a lattice is affine rather than homogeneous (a one-dimensional path has $|B(r)| = 2r + 1$ from an interior point, not r^1), which biases a naive fit on volume itself well below the true dimension at any computationally tractable radius. Shell size has no such additive offset (a path's shell is the exact constant 2; a 2D grid's shell is exactly $4r$), so the corrected estimator returns *exactly* 1.0 and 2.0 on these known lattices, not merely approximately.

2.3 From continuous trajectories to hypergraphs

A continuous physical trajectory (a set of points in phase space) is converted to a hypergraph via a k-nearest-neighbour proximity graph: each point becomes a node, and an edge connects each point to its k nearest neighbours (symmetrized). Graph distance on a sufficiently dense k-NN graph approximates geodesic distance on the underlying manifold, the standard justification for this class of dimension estimator (Grassberger & Procaccia, 1983; the same construction underlies Isomap-style manifold learning).

2.4 The traditional baseline

To make "better than the traditional approach" a checkable claim, the classical Grassberger–Procaccia correlation integral was implemented as an explicit reference method:

$$C(r) = \frac{2}{N(N-1)} \#\{(i,j) : i < j,\ |x_i - x_j| < r\}, \qquad D_2 = \frac{d \log C(r)}{d \log r}$$

Both this baseline and the shell-growth estimator are built on the same underlying primitive (`scipy.spatial.cKDTree`), so that any compute-cost comparison isolates the *algorithmic* difference (global pairwise scaling versus local graph growth) rather than an implementation-quality difference between a carefully optimized new method and a deliberately weak baseline.

2.5 Comparison methodology

For a given point cloud and known target dimension, each method's minimum point count for *stable* convergence is determined: the smallest n in a tested grid at which the estimate is within tolerance of the target *and remains* within tolerance at every larger n tested — guarding against a single lucky crossing of a still-moving estimate, the exact failure mode discovered in the initial chaotic-attractor results (Section 3). A "win" requires both methods to demonstrably converge; a method that never converges cannot be credited with needing "fewer" points, since that would reward failure to converge as though it were efficiency.

3 Initial Benchmark: Ten Known Physics Problems

The dimension estimator was applied to ten problems spanning exact analytic targets, a rigorous probabilistic theorem, and two literature values from classical chaos theory.

#	Problem	Known dimension	Source
1	Harmonic oscillator	1 (exact)	closed phase-space curve
2	Nonlinear pendulum (large amplitude)	1 (exact)	closed periodic orbit
3	Kepler two-body orbit ($e = 0.6$)	1 (exact)	closed ellipse
4	Mars vs. real JPL Horizons data	1 (exact)	smooth arc, real ephemeris
5	Quasi-periodic 2-torus (Lissajous)	2 (exact)	standard KAM-type result
6	Planar Brownian motion	2 (exact)	Taylor (1953) theorem
7	Restricted 3-body periodic orbit	1 (exact, if closure verified)	limit-cycle definition
8	Lorenz attractor	≈ 2.05	Grassberger & Procaccia (1983)
9	Rössler attractor	≈ 2.01	Grassberger & Procaccia (1983)
10	Driven damped pendulum (mode-locked)	1 (exact, if non-chaotic)	limit-cycle definition

Table 1: The ten benchmark problems and their independently-sourced target dimensions.

3.1 Headline result and why it is misleading

All ten problems passed their stated tolerance. An audit pass, followed by an independent verification of that audit (Section 3.2), established that this headline substantially overstates what was actually demonstrated:

- **Six of ten "passes" are structurally degenerate.** A k-NN graph built on any smooth closed curve is an exact circulant ring lattice, with a perfectly constant shell size at every radius. The estimator is mathematically incapable of returning anything other than dimension 1 with $r^2 = 1.0$ on such an input — the reported perfect fit is a sentinel meaning "the input had no variation to fit," not evidence that the method measured anything accurately. This affected problems 1, 2, 3, 4, 7, and 10.
- **Neither chaotic-attractor result was converged in point count.** A convergence sweep (varying the number of sampled points, all else held fixed) showed both the Lorenz and Rössler dimension estimates rising monotonically *through* their literature values and past them, still climbing at the largest point count tested. The reported "passing" errors (0.064 for Lorenz, 0.014 for Rössler) described where a still-moving quantity happened to be sampled, not a converged measurement.
- **The two remaining results (Lissajous torus, Brownian motion)** were the least-degenerate in the set, but a later correction found they had not been checked with the same convergence/sampling-grid discipline applied to the chaotic cases, and both showed comparable sensitivity to sampling-grid choice once checked.

3.2 Multi-level review chain

Ten independent agents each solved one problem and self-reported a result; all ten self-reports were later confirmed numerically correct. An independent audit reviewed all ten and correctly identified the structural problems above. A *second*, independent review was then applied specifically to the audit's own written analysis — and found three genuine errors in it: an over-generalized claim about the scope of a software defect (Section 4), a speculative reattribution that incorrectly overturned a separately-correct finding, and a factually incorrect claim about which sub-results had received a particular check, which had propped up one of the audit's two headline "success" verdicts without the same scrutiny applied elsewhere. All three errors are documented and corrected in the underlying benchmark record. This multi-level structure — workers, auditor, and an independent check on the auditor itself — is now standing practice for any result in this programme intended to be cited.

4 Software Defects Found and Corrected

Two independent, genuine defects in the estimator's own implementation were found as a direct consequence of running it against real problems rather than only the synthetic lattices used in its original unit tests.

4.1 $\mathbf{R}^2$ catastrophic cancellation

For a perfectly constant shell sequence, the total sum of squares in the R^2 computation should be exactly zero. Floating-point rounding in the intermediate mean computation left it at approximately 10^{-31} for specific fit lengths and shell values instead of exactly zero, causing the implementation's exact-zero comparison to take the general-case branch and divide by that residual — collapsing the reported R^2 to approximately zero by pure numerical noise, even though the underlying dimension estimate remained correct. This meant the estimator's own documented default settings returned an undefined result (`NaN`) for several ordinary lattices, including the exact configuration four of the ten benchmark problems used. The fix replaces the exact-zero comparison with a relative-scale tolerance; verified against every shell value found to trigger the defect (6, 17, 18 at the affected fit length) plus negative controls that were already correct, with two regression tests added.

4.2 Adjacency-reconstruction performance defect

A separate, unrelated performance defect was found while building the comparison harness: the hypergraph's adjacency structure was rebuilt from scratch, at $O(\text{edges})$ cost, on every single distance query, and the dimension estimator queried it once per radius per sampled node — compounding into thousands of redundant reconstructions for a realistic convergence sweep. A representative 3200-point sweep took 136 seconds before correction. The fix caches the adjacency structure per (immutable) hypergraph instance and restructures the dimension estimator to perform a single incremental breadth-first search collecting all radii's volumes in one pass, rather than recomputing from the source at every radius. The identical 3200-point sweep completed in 5.0 seconds after the fix — a $\sim 27\times$ improvement — with numerically *identical* results before and after, confirming the fix traded no correctness for speed. This performance improvement is the specific enabling condition for the systematic re-comparison reported in Section 5: the convergence sweeps it requires were not practically tractable before this fix.

4.3 Duplicate-point detection

A third improvement, not a bug fix but a robustness addition: sampling multiple periods of a closed orbit produces near-exact duplicate point clusters (separated by $\sim 10^{-9}$ in the cases observed), which silently corrupted three of the ten original benchmark results with severity ranging from a degraded fit quality to a complete failure to converge. The k-NN construction now detects duplicate *clusters* (not merely pairs, since a three-period orbit produces clusters of three) via a connected-components analysis on the within-threshold pair graph, calibrated against the point cloud's bounding-box diagonal rather than its local nearest-neighbour spacing — an initial attempt using the latter failed silently, because a duplicated point's nearest neighbour *is* its own duplicate, corrupting the very statistic meant to detect it. Verified on the exact reproduction of the original defect (300 points, each period-tripled at $\sim 10^{-9}$ offset): the corrected implementation raises an explicit error naming all 300 size-three clusters, and an automatic-cleaning option recovers exactly the original 300 points and the correct dimension of exactly 1.0.

5 Current Status of the Comparison to Traditional Methods

With the defects above corrected and a genuine traditional-method baseline implemented (Section 2.4), a systematic re-comparison across all ten benchmark problems was launched, targeting the programme's stated goal: a majority (at least eight of ten) of the problems demonstrating an advantage over the traditional correlation-sum method, defined by two explicit, checkable criteria rather than a single vague notion of "better":

1. **Compute savings**: more than 10% fewer sample points needed for equivalent, *stably converged* accuracy against the same tolerance.
2. **Density-robustness** (the concrete, checkable form of "singularity avoidance"): physical trajectories with a close approach or a slow/fast turning point (a near-primary passage in the restricted three-body problem, perihelion passage in an eccentric Kepler orbit, a pendulum's turning point) are sampled far more densely in time near that feature than elsewhere, even though the underlying curve is uniformly one-dimensional throughout. A *global* method that pools all pairwise distances into one statistic (the traditional correlation sum) can be measurably biased by such a density cluster; a *local* method that only ever examines each point's nearest neighbours (the shell-growth estimator) is not, by construction. This is tested directly by comparing each method's error against the known target on the natural, non-uniformly-sampled data.

This comparison is now complete, having passed through three independent review passes: self-report, audit, and an adversarial skeptic tasked with checking the audit itself (the same three-level chain used throughout this programme; see Section 3.2). One illustrative, manually-verified data point from the in-progress phase is retained for continuity: on a filled two-dimensional square (target dimension 2, tolerance 0.15), the traditional correlation-sum method reached stable convergence at 100 points while the shell-growth method required 400 (-300% relative point count, i.e. the opposite of a poly-algebraic win) — consistent with the final result below, which finds no advantage on non-degenerate two-dimensional targets either.

Final verified count: 4 of 10 (goal not met). The programme's stated goal was at least eight of ten problems. Four problems (the nonlinear pendulum, the driven pendulum, a Kepler orbit, and a Mars-like orbit) show a genuine compute-savings advantage, each converging at 64 points against the traditional method's 256 (savings 0.75), robust across every k and fitting-window swept. Zero of four density-robustness claims survived a uniform-resampling control: on every system tested with a genuine density feature (the nonlinear pendulum, the driven pendulum, a Kepler orbit, the restricted three-body problem), the traditional method was equally or slightly *more* accurate on the density-varying sample than on a uniform control, the opposite of the hypothesised effect. **All four verified wins share one underlying mechanism**: each is a smooth, closed (or near-closed) one-dimensional orbit, which puts the k-nearest-neighbour graph into an exact ring-lattice regime where the shell-growth estimator's fit is degenerate (Section 6) but numerically correct, and converges from far fewer points than a global correlation sum needs to escape its own fitting-window bias. The suite contains no second, structurally different compute-savings win.

One result deserves separate mention because the win criterion is structurally unable to score it: on two non-degenerate, genuinely two-dimensional targets in the suite (a quasiperiodic torus and a planar Brownian path), the shell-growth estimator is accurate to within 0.03–0.05 of the true value with zero degenerate fits, while the traditional correlation sum converges confidently (fit quality $R^2 > 0.998$) to a value wrong by 0.36–0.61. The traditional method never stably converges on these two targets at all, so the "fewer points to converge" criterion — correctly, per this programme's standing rules — cannot credit a comparison against a method that never converges. This is nonetheless the strongest evidence in the whole benchmark that the estimator captures something real beyond the degenerate ring-lattice case, and the next round's headline question should be accuracy on non-degenerate measures rather than points-to-converge.

The density-robustness criterion is retired in its current form. After four independent attempts, all null or sign-inverted under a proper control, it produced no supporting evidence in this benchmark and is not carried forward without a mandatory uniform-resample control arm and a target system where the estimator is not structurally pinned to a single answer. The full derivation, including three levels of independent re-verification of every reported number, is in `docs/POLY_ALGEBRAIC_BENCHMARK.md`, Sections 9–10.

6 Limitations

- **The estimator has no freedom to be inaccurate in the degenerate regime.** Any smooth, densely and evenly sampled closed curve will read as dimension 1 with a perfect (but uninformative) fit quality, regardless of the method's actual accuracy. A `degenerate` flag now distinguishes this case explicitly in the software, but any result drawn from problems 1, 2, 3, 4, 7, or 10 of the original benchmark should be read as confirming the pipeline is correctly wired, not as evidence of measurement accuracy.

- **Fractal (non-integer) dimension targets need 2–4× more points to stabilise than the traditional method**, per the completed comparison of Section 5: on both chaotic attractors tested, the shell-growth estimator converges more slowly than the traditional correlation sum, though it lands closer to the true value once it does. A distinct, still-open estimator defect (near-constant but not exactly-constant shell sequences collapsing the fit-quality statistic unreliably) is tracked as N8 in `docs/POLY_ALGEBRAIC_BENCHMARK.md`.

- **The k-nearest-neighbour construction remains an approximation**: graph distance approximates but does not equal geodesic distance on the underlying manifold, and this approximation degrades near a manifold's boundary (documented and pinned by a regression test showing measurable underestimation at the corner of a sampled cube) and is sensitive to the choice of k and to local sample density.

- **This work does not validate Wolfram's physical hypothesis.** Everything reported here is a statement about the behaviour of a specific graph-theoretic algorithm on specific finite point clouds — Tier B by this programme's epistemic convention. No claim is made or implied about whether hypergraph rewriting is a correct fundamental description of physical spacetime; that remains Tier C (labelled conjecture, never load-bearing) throughout.

7 Discussion

The most consequential finding of this stage is procedural rather than purely technical: a benchmark against real, independently-sourced targets found real defects (Section 4) that the estimator's own unit tests, exercised only against synthetic lattices designed to be easy, could not have found. Equally consequential is that the *review of the benchmark* itself required its own independent check before its conclusions could be trusted — an audit that correctly identified real problems in the original work was not, on that basis alone, exempt from containing real problems of its own. Both observations are recorded as general lessons for this programme (see the accompanying lessons-learned document) and are now built into the standing workflow pattern used for any result in this repository intended to be cited: measurement, audit, and an independent check on the audit, in that order, with none of the three treated as sufficient on its own.

8 Conclusion and Next Steps

Phase 2, Stage 2 of this programme has produced: (i) a working, tested implementation of two of the four brainstormed hypergraph-physics formalization concepts; (ii) a ten-problem benchmark against independently-sourced physics targets, whose headline result required substantial correction under audit but whose underlying software defects were genuinely found and genuinely fixed; (iii) a real classical baseline and comparison methodology, replacing what would otherwise be an unfalsifiable claim of superiority with two specific, checkable criteria; and (iv) a completed, three-times-independently-verified systematic comparison across all ten problems, reported in Section 5. The programme's stated goal (at least eight of ten problems showing an advantage) was not met: the final, corrected count is four of ten, all four sharing one underlying mechanism (a smooth closed one-dimensional orbit's ring-lattice degeneracy), and no density-robustness advantage was demonstrated anywhere in the suite. The most informative result the benchmark produced — accuracy on two non-degenerate two-dimensional targets where the traditional method never converges — is structurally outside what the stated win criterion can score, and is the recommended headline question for the next round rather than points-to-converge. This outcome is reported plainly, as the programme's standing methodology requires: an honest negative or partial result is treated as a completed piece of work, not as a phase awaiting a better number.

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Internal references: `docs/HYPERGRAPH_NOTES.md`, `docs/POLY_ALGEBRAIC_BENCHMARK.md`, `docs/LL.md`, `src/socrates/hypergraph/`.